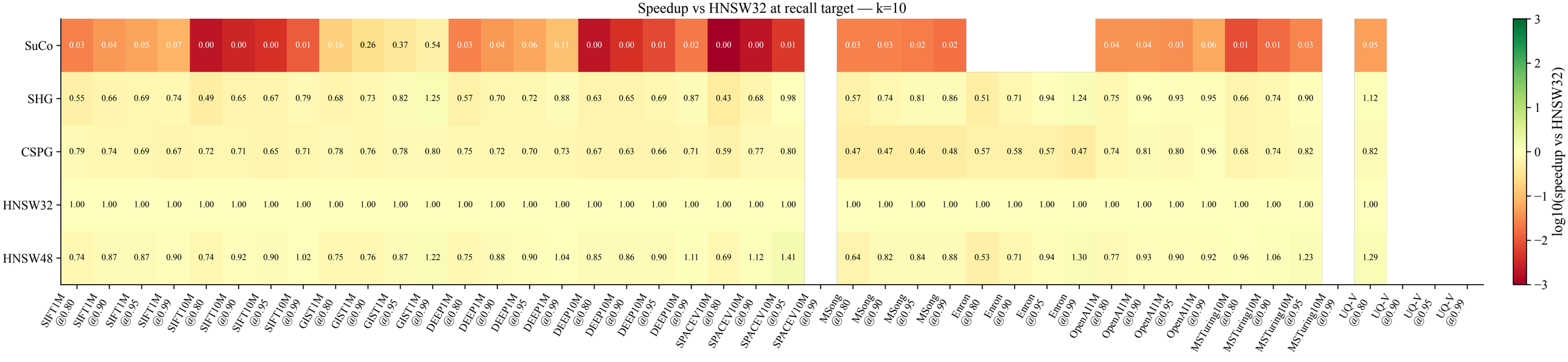
Speedup vs HNSW32 at recall target — k=10
SuCo
SHG
CSPG
HNSW32
HNSW48
0.03 0.04 0.05 0.07 0.00 0.00 0.00 0.01 0.16 0.26 0.37 0.54 0.03 0.04 0.06 0.11 0.00 0.00 0.01 0.02 0.00 0.00 0.01 0.03 0.03 0.02 0.02 0.04 0.04 0.03 0.06 0.01 0.01 0.03 0.05
0.55 0.66 0.69 0.74 0.49 0.65 0.67 0.79 0.68 0.73 0.82 1.25 0.57 0.70 0.72 0.88 0.63 0.65 0.69 0.87 0.43 0.68 0.98 0.57 0.74 0.81 0.86 0.51 0.71 0.94 1.24 0.75 0.96 0.93 0.95 0.66 0.74 0.90 1.12
0.79 0.74 0.69 0.67 0.72 0.71 0.65 0.71 0.78 0.76 0.78 0.80 0.75 0.72 0.70 0.73 0.67 0.63 0.66 0.71 0.59 0.77 0.80 0.47 0.47 0.46 0.48 0.57 0.58 0.57 0.47 0.74 0.81 0.80 0.96 0.68 0.74 0.82 0.82
1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00 1.00
0.74 0.87 0.87 0.90 0.74 0.92 0.90 1.02 0.75 0.76 0.87 1.22 0.75 0.88 0.90 1.04 0.85 0.86 0.90 1.11 0.69 1.12 1.41 0.64 0.82 0.84 0.88 0.53 0.71 0.94 1.30 0.77 0.93 0.90 0.92 0.96 1.06 1.23 1.29
SIFT1M @0.80
SIFT1M @0.90
SIFT1M @0.95
SIFT1M @0.99
SIFT10M @0.80
SIFT10M @0.90
SIFT10M @0.95
SIFT10M @0.99
GIST1M @0.80
GIST1M @0.90
GIST1M @0.95
GIST1M @0.99
DEEP1M @0.80
DEEP1M @0.90
DEEP1M @0.95
DEEP1M @0.99
DEEP10M @0.80
DEEP10M @0.90
DEEP10M @0.95
DEEP10M @0.99
SPACEV10M @0.80
SPACEV10M @0.90
SPACEV10M @0.95
SPACEV10M @0.99
MSong @0.80
MSong @0.90
MSong @0.95
MSong @0.99
Enron @0.80
Enron @0.90
Enron @0.95
Enron @0.99
OpenAI1M @0.80
OpenAI1M @0.90
OpenAI1M @0.95
OpenAI1M @0.99
MSTuring10M @0.80
MSTuring10M @0.90
MSTuring10M @0.95
MSTuring10M @0.99
UQ-V @0.80
UQ-V @0.90
UQ-V @0.95
UQ-V @0.99
log10(speedup vs HNSW32)
3
2
1
0
−1
−2
−3